

# Robin T Carpenter

West Lafayette, IN • 47906 • USA  
920 - 382 - 5020

Email: [carpenterofcomputing@gmail.com](mailto:carpenterofcomputing@gmail.com)  
[www.linkedin.com/in/robin-carpenter-of-computing](https://www.linkedin.com/in/robin-carpenter-of-computing)  
Personal website: [taunoneutrino.github.io](https://taunoneutrino.github.io)

> ----- <

## – RESEARCH EXPERTISE

### *Skills and academic experience*

- Data analysis, Python & R (**7 yrs.**)
- Tech. documentation (**5 yrs.**)
- Script & backend development (**4 yrs.**)
- Presentation & scientific comm. (**4 yrs.**)
- Project management and resource leadership (**4 yrs.**)
- Supercomputing & big data (**3 yrs.**)
- Machine learning (**3 yrs.**)
- Linear algebra (**3 yrs.**)
- Metrology & Optics (**Oscilloscopes, multimeters, microscopes 2 yr.**)
- ESD Cleanroom etiquette (**1 yr.**)

*Computing Techniques:* **Numerical, Monte Carlo, quantum and simulated annealing, machine learning, vectorizing, tensor networks**

*Applications (internships/projects):* **AutoCAD, Matlab, PowerBI, ArcGIS, Microsoft Suite.**

## – EDUCATION

*Purdue University -( 610 Purdue Mall, West Lafayette IN )*

[ 2022 ] • **Applications in Data Sciences** Certificate

[ 2023 ] • Minors in **Planetary Science, Spanish for the Professions, Astronomy**

[ 2025 ] • **BS in Physics**, 3.37 GPA.

*Topics in excellence:* **Optics, Astronomy, Particle Physics, Computation**

## – RESEARCH EXPERIENCE

### *Semester Projects*

[ 2021 ] • Succeeded in **enhanced background subtraction** from 1 TB of dark matter scintillation data provided by XENON 1T.

[ 2022 ] • Reported gaps in Alerce, MARS, TNS for transient photometric measurements using **API scraping**

[ 2023 ] • Contributed with a team of developers within a **Django web backend** for the Mara Elephant Project

[ 2024 ] • Batched simulated data for the Jung Research Group in quantum computing benchmarking, under a **supercomputing cluster**

### *Upper Level Projects*

[ 5/2023 - 8/2023 ] • **Image Metrology Intern** - ( *Technische Universität Dortmund, Dortmund Germany* )

Completed project under the ATLAS Phase-II project: developed enhanced angle and edge detection of micron scale silicon detectors under supervision. Assisted in improving resolution tenfold

[ 5/2025 – 8/2025 ] • **Quantum Machine Learning Research Member** - ( *Purdue University D. of Physics and Astronomy* )

Scripted 4 methods of measuring non-stabilizerness of quantum states, concerning Second Rényi Entropy and Mana. Currently applying methods to novel QVE ansatz.

### **– TEACHING EXPERIENCE**

#### *Undergraduate Teaching Assistant*

[ 2023 – 2025 ] • ( *The Data Mine Physics Cohort, Quantum Computing group* ) Personally taught students essentials of quantum information science, organized presentations and group projects, and prepared students for presenting posters at symposiums.

### **– SERVICE/LEADERSHIP**

- Purdue Climate Action Committee (*1 yr.*)
- Purdue Bands and Orchestras (*4 yrs.*)

### **– LANGUAGES**

[ *Fluent* ] • English  
[ *Intermed.* ] • Spanish  
[ *Beginner* ] • Chinese